Discrete Mathematics Spring 2026

Homework 9: Induction and Countability

Due Sunday April 12 @11:59:00pm Eastern

Show all your work to receive full credit

1. (25 points) **Proof by induction**

   Assume n is a positive integer. Use induction to prove the following:

   $$1 + 5 + 9 + \cdots + (4n - 3) = 2n^2 - n$$

2. (25 points) **Proof by smallest counter example**

   Let n be a positive integer. Prove by **smallest counter example** that:

   $$\forall n > 0 \quad \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} = \frac{2^n - 1}{2^n}$$

3. (25 points) **Strong Induction**

   Prove the following statement by strong induction:

   There exist non-negative integers x, y such that $n = 3x + 7y$ for all $n \geq 12$.

4. (25 points) **Countable sets**

   Prove by **constructing an explicit enumeration** that the set $\mathbb{Z} \times \mathbb{Z}$ is countable.